



Notes

Jun 25, 2026

Home Depot Discussion

Invited [Vaibhav Singh](#) [Sameer Shaik](#) [Asit Srivastava](#) [Avinash Patel](#)
[Harish Kumar S](#) [Snehashish Debnath](#) [Fatemeh Teimury](#) [Saivijaay VK](#)

Attachments [📎 Home Depot Discussion](#)

Meeting records [📄 Transcript](#)

Summary

The team discussed onboarding new members, reviewed technical pipeline improvements for object detection, and planned upcoming diagnostic assessments.

Team Integration and Cadence

The team formalized onboarding processes for new members and established a mandatory daily standup meeting to ensure alignment. Leadership recognized internal growth with the assignment of new project oversight roles.

Pipeline Development and Optimization

Performance reached 92 percent accuracy following pipeline modifications and the integration of automated machine learning operations. Technical discussions addressed persistent issues with product detection, label orientation, and data annotation requirements.

Strategic Future Direction

The team shifted focus toward diagnosing system failure points in object detection and optical character recognition workflows. Future priorities

include multiframe integration and refined data annotation strategies to enhance system reliability.

Decisions

Aligned

- **MLOps pipeline requirement set** The MLOps pipeline development is established as a mandatory project requirement rather than an optional initiative.
- **Product detection pipeline strategy refined** The product detection strategy is defined to prioritize identifying at least one valid product per price tag to feed the embedding pipeline, rather than focusing on exhaustive cluster product detection.

We've **updated the Decisions section** using your feedback.

Let us know what you think: [Helpful](#) or [Not Helpful](#)

Next steps

- ☐ [Asit Srivastava] Onboard New Team Member: Assign ID and coordinate onboarding process for the new team member, Sneish, into Home Depot.
- ☐ [Vaibhav Singh] Send Email Addresses: Provide email addresses for the new team members to Asit Srivastava to facilitate the onboarding process.
- ☐ [Vaibhav Singh] Review Project Status: Meet with Sneish to understand the progress and extent of work completed on his assigned project.
- ☐ [Sameer Shaik] Deploy MLOps Pipeline: Establish and deploy the MLOps pipeline on the Home Depot machine.
- ☐ [Sameer Shaik] Create Presentation: Create a basic presentation summarizing current project status, planning, and next steps for the MLOps work.

- ☐ [Samir Shaik, Fatemeh Teimury] Provide System Access: Provide VPN access and GPU access to Avinash Patel.
- ☐ [Vaibhav Singh] Request Re-annotation: Request that the team re-annotate images specifically for cluster products and individual products to improve detection performance.
- ☐ [Fatemeh Teimury] Gather and Annotate Data: Retrieve 9000 pending images from Snare and Harish. Pre-annotate these images to prepare them for Sumit.
- ☐ [Fatemeh Teimury] Send Documentation: Forward the shared documentation to Avinash. Provide the materials requested for his onboarding.
- ☐ [Avinash Patel] Review Project Materials: Read the provided documentation. Explore the software to familiarize yourself with the current project status.
- ☐ [Fatemeh Teimury] Refine Detection Logic: Assess the effect of modifying cluster product definitions on detection performance. Test whether merging cluster products with standard products improves results.
- ☐ [The group] Complete Task Lists: Update individual task lists. Ensure details are comprehensive for the end of month presentation.
- ☐ [Sameer Shaik, Fatemeh Teimury] Setup Technical Access: Assist Avinash with necessary technical access. Help with ticket raising and VPN or on premises setup.
- ☐ [Vaibhav Singh] Develop Verification Script: Create a script to analyze failure modes in the single line verification job. Use the script to pinpoint issues in segmentation and object detection pipelines.
- ☐ [Harish Kumar S] Provide Image Samples: Send sample images to Avinash once machine access is granted. Provide the files required to evaluate the current single line detection approach.
- ☐ [Vaibhav Singh] Distribute Project Files: Share the project zip file and results documents with Avinash. Explain the contents of these files during the next discussion.

Details

- **Team Introductions and Onboarding:** Vaibhav Singh welcomed Avinash Patel to the team and requested Asit Srivastava's assistance in initiating the formal onboarding and ID creation process for Avinash Patel at Home Depot ([00:03:31](#)).

Saivijaay VK, who was acknowledged for significant contributions behind the scenes, was congratulated for taking on a leadership role for a new team of 10 to 12 people, though they will remain in the group ([00:07:22](#)).

- **Communication and Meeting Cadence:** Vaibhav Singh emphasized the need for better communication regarding object detection work and established a mandatory daily stand-up meeting at 7:30 p.m. to ensure alignment. Additionally, Vaibhav Singh plans to meet with Sneha, who is currently on sick leave, to assess the status of an exclusively owned project and determine whether to delegate the work or assume responsibility for it personally ([00:10:35](#)).
- **Onboarding Capacity and Timelines:** Avinash Patel is currently finishing work on a Presto project for Home Depot but noted that the majority of tasks are nearing completion ([00:11:47](#)). Asit Srivastava suggested coordinating with Sahil to manage Avinash Patel's capacity effectively and estimated that the official ID and onboarding process would take approximately two days ([00:12:43](#)).
- **Single Line Pipeline Progress:** Vaibhav Singh reported a 5% accuracy boost in the single line pipeline, which improved from 87% to 92%, and confirmed that the team is preparing for pilot deployments. The team is also planning to integrate multi-frame information into the pipeline, with Harish Kumar S coordinating those efforts with Devon ([00:14:52](#)) ([00:16:52](#)).
- **Visual Embeddings Development:** The team discussed the development of a visual embeddings pipeline, which is currently paused but remains a priority for complementing existing SKU matching processes. This approach aims to assist in resolving edge cases that the current SKU matching process cannot handle ([00:16:52](#)).
- **Management and Stakeholder Interaction:** Vaibhav Singh provided an overview of the management hierarchy, noting that individuals such as Arun, Deepak, and Todd attend meetings occasionally, while Eldar joins rarely ([00:17:56](#)). Vaibhav Singh clarified that despite these interactions, the team's standard internal stand-up will continue as scheduled ([00:18:45](#)).
- **MLOps Pipeline Implementation:** Sameer Shaik outlined the development of an MLOps pipeline on GCP that currently utilizes the YOLO X model for object detection ([00:20:47](#)). This workflow includes uploading data to Google data buckets, with training, logging, and monitoring automated through Vertex AI pipelines ([00:22:00](#)). Key features implemented include dataset visualization for tracking classes and the use of Vertex AI experiments to monitor hyperparameters and artifacts ([00:23:07](#)).

- **Automation and Deployment Strategy:** Sameer Shaik detailed the current progress on automating MLOps, which includes building Docker containers, pushing images to a registry, and working toward deploying to Vertx endpoints ([00:24:06](#)). The team discussed future plans to implement event triggers via Pub/Sub or Eventarc to automate training whenever changes occur in the Google data bucket ([00:25:22](#)).
- **Managing Stakeholder Expectations:** Vaibhav Singh advised the team to prepare for potential pushback from existing groups accustomed to current workflows, suggesting that the team members should prepare thorough answers to navigate these interpersonal and technical challenges ([00:31:23](#)).
- **Access and Technical Troubleshooting:** The team discussed resolving access issues, specifically regarding VPN and GPU access for Avinash Patel ([00:32:40](#)). Fatemeh Teimury reported a persistent inability to access a VM at IP address 172.16.20.108, and the team committed to investigating and resolving this connection issue ([00:33:38](#)).
- **Product Detection and Segmentation:** Fatemeh Teimury explained the current focus on YOLO-based detection and segmentation models to extract product crops, which will serve as input for the secondary visual embedding pipeline ([00:35:17](#)). The team aims to detect products and cluster products to optimize the supply chain, although current model performance is limited by inconsistent data annotations ([00:37:30](#)).
- **Data Annotation Strategy:** Vaibhav Singh and Fatemeh Teimury identified a requirement to re-annotate approximately 9,000 images to resolve discrepancies in how "product" and "cluster product" are defined ([00:47:24](#)). They explored strategies to improve detection, including merging cluster products with individual products or handling them through a stage-based network ([00:44:29](#)) ([00:52:43](#)).
- **Anomaly Detection for Empty Shelves:** Vaibhav Singh discussed utilizing anomaly detection, such as anomaly clipping, to identify empty shelf spaces ([00:48:06](#)). The team agreed that while current models perform well on price tags and products, they will continue to refine the approach for detecting empty regions provided the primary requirement of identifying price tags and products is met ([00:48:58](#)).
- **Single Line Detection Challenges:** Harish Kumar S described the Single Line Detection problem, highlighting the functional distinction between the "selling area" where customers retrieve items and the "overhead area" used for stock

(00:53:41). The team discussed the limitations of Google OCR, including its sensitivity and latency, and the potential need for a custom digit detection model (00:56:28).

- **Orientation and Rotation Correction:** Harish Kumar S and Vaibhav Singh discussed a critical difficulty with single line SKU labels, which lack the anchor points used in multi-line labels to correct for rotation (00:56:28) (01:01:02). The team concluded that they need to evaluate the prevalence of incorrectly oriented labels and explore methods for orientation correction, potentially through new OCR techniques or specialized digit detection algorithms (00:58:08).
- **Pipeline Modification and Performance:** Harish Kumar S describes a modified two-stage pipeline where the first stage involves detecting single-line labels from an overhead image and organizing them into a horizontal strip for segmentation (01:02:12). This segmentation phase maps polygon points back to the original image to exclude noise, producing a masked image used by Google OCR to extract skew values. Harish Kumar S reports that this process resulted in a 5% performance boost, bringing current accuracy to 91.8% (01:03:49).
- **OCR Extraction and Data Filtering:** Regarding the OCR process, Harish Kumar S explains that while Google OCR successfully handles labels with 90-degree orientations, it fails to process upside-down labels. The team aims to transition to a task-based OCR to improve performance (01:03:49). To ensure data quality, the team applies regular expression patterns to the text annotations after the OCR stage, filtering out noise and false positives to ensure the final skew values are either six or 10 digits long (01:05:17).
- **Access to Codebase and Data:** Avinash Patel asks about obtaining access to the project's codebase to facilitate exploration. Harish Kumar S clarifies that once Avinash Patel gains access to the machine, they will have access to the common data repository where images are stored. In the interim, Harish Kumar S agrees to share the necessary images with Avinash Patel (01:07:11).
- **Team Updates and Upcoming Work:** Vaibhav Singh updates the team on their progress, noting a transition from three weeks of data annotation to a new project focused on identifying failure points in the Google OCR, segmentation, and object detection systems. Although current accuracy is at 92%, Vaibhav Singh intends to develop a script to diagnose specific causes of failure. Additionally, Vaibhav Singh mentions that the team, including Harish Kumar S, will begin work on multiframe integration the following day (01:08:29).

- **Task Documentation and Reporting:** Vaibhav Singh requests that all team members, including Sameer Shaik, Harish Kumar S, and Fatemeh Teimury, fill out their task lists in detail to facilitate an end-of-month presentation. Vaibhav Singh emphasizes the necessity of these detailed updates for reporting purposes ([01:09:29](#)).
- **Administrative Onboarding and Knowledge Transfer:** Vaibhav Singh instructs Avinash Patel to coordinate with Sameer Shaik and Fatemeh Teimury regarding ticket raising, on-premises access, and VPN requirements. Vaibhav Singh also confirms they will add Avinash Patel to the Home Depot group. To assist with the transition, Vaibhav Singh agrees to share relevant documentation, a results document, and a zip file, with a plan to explain the contents of the zip file to Avinash Patel the following day ([01:10:35](#)).

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Transcript

Jun 25, 2026

Home Depot Discussion - Transcript

00:03:31

Harish Kumar S: Good morning everyone.

Fatemeh Teimury: Where is it?

Harish Kumar S: Morning. So WBO will be joining in five minutes. We can continue.

Asit Srivastava: Hey guys,

Harish Kumar S: Nice. How are you

Avinash Patel: Let's

Asit Srivastava: I for

Harish Kumar S: doing?

Fatemeh Teimury: Hello.

Vaibhav Singh: Hey,

Asit Srivastava: Hey baby, how are you?

Vaibhav Singh: very bad. Very tense. Very bad.

Asit Srivastava: What's happening? What's what's the context? I'm just curious.

Vaibhav Singh: Uh nothing. So this meeting is just for So Avin is going to join our team. is yeah Hinash is there uh so is not there okay no worries so I will join uh what I this is just a small get together about the team who are there in the team that's it and EB asked me to just have everyone so everyone knows who is working so couple of guys are here couple of guys are in the US and one is in Canada everyone just knows about it and one other task uh is uh Sneish is not going to join today

00:06:15

Asit Srivastava: are

Vaibhav Singh: But him and a uh ID has to be created. So deeper. So whatever process has to be started can you help with that? Ain. Yeah. Ainina and Dnath.

Asit Srivastava: having.

Vaibhav Singh: I'll send their emails with you to you as well. But uh these guys have to be onboarded in Home Depot. Okay.

Asit Srivastava: Okay.

Vaibhav Singh: I'm not sure what's the official way to go for so I'm I'm deferring this to you.

Asit Srivastava: Sure.

Vaibhav Singh: Uh yeah, Vin.

Asit Srivastava: Sure.

Avinash Patel: Hello. How are

Vaibhav Singh: Hey.

Avinash Patel: you?

Vaibhav Singh: Uh so you got a chance to meet with Samir yesterday?

Avinash Patel: Uh, briefly. I had to go to a different meeting, so I we talked just a little bit,

Vaibhav Singh: No,

Avinash Patel: but um uh no,

Vaibhav Singh: no worry.

Avinash Patel: not like a full conversation.

Vaibhav Singh: No worries. Uh so uh just to give her so there there's Harish she already saw yesterday but Harish is in the meeting and Fatima is in the meeting as well.

00:07:22

Vaibhav Singh: So she is in PST time zone. Hello Fatima.

Fatemeh Teimury: Hi everyone, nice meeting you.

Vaibhav Singh: Hello.

Fatemeh Teimury: Welcome to the

Vaibhav Singh: Yeah.

Avinash Patel: Thank you.

Asit Srivastava: Hello.

Vaibhav Singh: Hello.

Avinash Patel: Thank you.

Fatemeh Teimury: team.

Vaibhav Singh: So Harish is here and we have one more. We had one more person. Siju okay Sij is also there but uh probably he won't be joining going forward he was not officially onboarded in Home Depot but he used to do all the work in the background him andish used to do all the work in the background uh right now Saij is going to lead another team so he's being he'll be busy over there and yeah okay he's joined And yes Sai and one another purpose for this meeting one another purpose for this meeting was to just congratulate Sai Vijj. So he's going to lead a team of 10 or 12 people right now and just want to

Fatemeh Teimury: And

Vaibhav Singh: appreciate all the work he has done. So if you have some work or have some doubt you can still irritate him.

00:08:25

Fatemeh Teimury: then

Vaibhav Singh: Uh there's a full there's no blockers over there. You can message him at any time. Okay. So, uh he's going to start working for NIQ and that's why he's he's still he'll still

Fatemeh Teimury: what?

Vaibhav Singh: be there in the group. I won't remove him from the group, but probably going to work as much as before. So, yeah, just a little appreciation. Thank you all for all the works and I know Fatima and Samir has relied a lot on

Saivijaay VK: Thanks very much.

Vaibhav Singh: you. Samir is not here.

Fatemeh Teimury: Thanks for all the help.

Vaibhav Singh: Okay.

Asit Srivastava: Come on. I'm

Vaibhav Singh: Okay. Okay.

Saivijaay VK: Thanks. Like I'm seeing a lot of new faces

Vaibhav Singh: Yeah. Yeah.

Asit Srivastava: outside.

Vaibhav Singh: So, sorry. As a Sith has joined recently.

Saivijaay VK: actually.

Vaibhav Singh: Avin has joined recently. So, I'll just Ait is our customer success for Home Depot right now. And Avin is going to be a new joiner for this team and he's in the same time zone.

00:09:31

Vaibhav Singh: He's with uh in the Tampa, right? I mean, yeah. So,

Avinash Patel: Yeah.

Vaibhav Singh: he's sitting from Tampa from the US office and Yeah. So, he'll be taking over some of the work, right?

Asit Srivastava: She's also engaged in another project at home depot.

Vaibhav Singh: Oh,

Asit Srivastava: So he's doing a little crisis.

Avinash Patel: Yeah, this is this is my second endeavor into Home Depot. It's

Asit Srivastava: Yeah. Yeah.

Vaibhav Singh: okay. Okay.

Asit Srivastava: Yeah.

Vaibhav Singh: Okay.

Avinash Patel: fun.

Vaibhav Singh: So,

Asit Srivastava: So um so just um so did Sahil connect with you about engaging Ain?
How did you uh end up

Vaibhav Singh: uh AB AB Aul at Bun Atul

Asit Srivastava: engaging?

Vaibhav Singh: Bunul uh just mentioned just connect with him.

Asit Srivastava: Okay.

Vaibhav Singh: So I had a good chat about half an hour with him a yesterday right so I just wanted to connect over here so everyone knows is this I think there's no other content for this meeting just

Avinash Patel: Yep.

Vaibhav Singh: everyone knows who all are there and going forward I have set up a internal meeting at 7:30 p.m.

00:10:35

Vaibhav Singh: I'm not sure what is that in EST or PST but if that time doesn't work let me know okay because we need right now the problem that I'm seeing is there's a lack of communication between team members about the same kind of stuff that we have been doing people are working in object detection but they are using applying some ideas but that ideas are not getting communicated with other people's other works so that stand up you'll have every day okay without fail it's required and another thing is so right now sne is on leave he is sick and because of that he's he was working on one of the projects now because he was the sole person owning that project no one knows the extent of work that has been done in that project and that is was one of our main stories going for in this for this Oh, so that means all the work has been stopped. Tomorrow I have a meeting with him. He'll start working from tomorrow. But I have a meeting with him to understand what has been done.

00:11:47

Vaibhav Singh: Who can I can delegate someone else to work or I can take over the work itself myself. Okay. So that's it. Uh regarding the project, Ainina you'll be starting out with. So uh just just a quick understanding uh if you're working on another project for the home depot as well. So that is with DH many sir.

Avinash Patel: Yeah. Yeah. It's the Presto

Asit Srivastava: Yeah.

Vaibhav Singh: Okay, nice. Uh,

Avinash Patel: project.

Vaibhav Singh: so just to understand your hour, work hours, how much how will you be able to manage? So is it actively are you contributing over there or uh what's what's going

Avinash Patel: Yeah, we're we're demoing some like a good part of it soon.

Vaibhav Singh: on?

Avinash Patel: So, I feel like the majority of the work is almost like on its way out hopefully. Like it's it's not going to be like a m it's not going to be a lot of uh like ongoing stuff for an extended period of

Asit Srivastava: Yeah, there's other person.

00:12:43

Asit Srivastava: So,

Avinash Patel: time

Asit Srivastava: u who should take over? Okay, so let me uh hopefully I'll connect with Sahil today. Uh man, I I'm thinking that Aina should free up uh because it was it was pretty much a

Vaibhav Singh: That's good. That's good. Uh

Asit Srivastava: oneperson work that has been happening for some time and that person in the US has not been panning out and the work has been extremely delayed. Okay. So, so um Ainash and other person were trying to pitch in and and kind of uh do some replacement work. So that's where we are. But I think his band would should pre I'm assuming that a be connected with Sahil and and that's how he's engaging Ainach. But let me also chat with Sahil because I was going to ask today the customer to on

board him for his project. Uh so I'm glad this meeting happened. So I'll I'll put a pause on that for right now. And it's still Home Depot but it'll be it would have been through a different um stakeholder.

00:13:45

Asit Srivastava: Um but anyway uh but I think it should should work

Vaibhav Singh: Yeah.

Asit Srivastava: out.

Vaibhav Singh: Okay. Yeah, sounds good. uh how I I don't have a rough estimate but how much time does it get to get an ID and a proper onboarding

Asit Srivastava: Weber for uh this project that you guys engaged in. Uh I think u Sanjay do you know Sanjay?

Vaibhav Singh: done

Asit Srivastava: Sanjay Ma he's the one who will do it. Okay. And I think it's pretty quick. Uh once we send a request to it it'll get done I'm assuming in a couple of days. Okay. Um

Vaibhav Singh: couple of days. Okay. Yeah, sure. Sounds good. Okay. Um, so Vin, just a final formal hello to the team. Welcome to the wing. We are excited and all the projects you can help us with. And next thing either we can have in the next meeting or what we can do is each one of them her Sam Harish can tell Spatima and Samir can just tell you what's the existing project they are working on.

00:14:52

Vaibhav Singh: Okay, Asset, if you want to stay back, you can stay back. But we are just going to talk about the project right

Asit Srivastava: I I'll stay I'll stay back.

Vaibhav Singh: now.

Asit Srivastava: I was going to ask you uh get a general sense of where we are. What's the latest?

Vaibhav Singh: Oh yeah,

Asit Srivastava: I mean I after you also with Shamir I had some sense but any

Vaibhav Singh: yeah, yes, yes, yes.

Asit Srivastava: developments after that I don't have good sense on you know where we

Vaibhav Singh: Yes, yes, yes.

Asit Srivastava: are.

Vaibhav Singh: So one uh good news is the single line pipeline that we were building we have more than 5% boost on their existing pipeline that is 87 to 92. they have decided to start piloting in a

Asit Srivastava: Okay,

Saivijaay VK: Nice.

Vaibhav Singh: pilot deployments and they will will be properly start working on the multi-frame integration into our single line pipeline.

Asit Srivastava: I see. Okay.

Vaibhav Singh: So multi multire was the multiple frames per bay and we'll start uh I think he has set up a call or he will set up a call Harish with teon and we'll start working on that.

00:15:58

Vaibhav Singh: So good news regarding that. I've been working a lot about that and there is still significant work pending in on that side but at least for it's good for now. It's good to go for now

Asit Srivastava: The existing pipeline was it a ML pipeline or was it a computer vision

Vaibhav Singh: computer vision.

Asit Srivastava: pipeline

Vaibhav Singh: So it was very easy pipeline but there were some problems in that pipeline and even though there were problems they were getting 87% accuracy that we got to

Asit Srivastava: Okay.

Vaibhav Singh: know very late and we were not expecting that and after we got to know so we did

Asit Srivastava: Okay. Okay.

Vaibhav Singh: some more modifications and now the accuracy is at a good stage that they

Asit Srivastava: Okay.

Vaibhav Singh: are u they want to go forward with the

Saivijaay VK: Still

Vaibhav Singh: deployment at least the pilot deployment

Asit Srivastava: So, they already gone through some testing in in a store too.

Vaibhav Singh: uh they will they will start. So that ticket is also on us.

Saivijaay VK: what?

00:16:52

Vaibhav Singh: But uh for me the past 3 4 days again I was busy with some annotation work for some other ticket and that has just got completed right now and from tomorrow or

Asit Srivastava: Okay.

Vaibhav Singh: today itself we'll start with the how to the deployment integration stuff with Devon. Basically we'll work with Devon a lot going forward.

Asit Srivastava: Okay.

Vaibhav Singh: Uh yeah so that's a good news visual embeddings work has started but it's on pause it's it's it's there's hope over there uh Sara hopes that visual embeddings will complement the their SQ matching existing

Asit Srivastava: Okay.

Vaibhav Singh: SQ matching process and it will help with the edge cases that SQ matching is not able

Asit Srivastava: Okay.

Vaibhav Singh: to properly uh fix or determine and

Asit Srivastava: Okay.

Vaibhav Singh: uh Yep.

Asit Srivastava: Okay.

Vaibhav Singh: Good stuff going on. But let's see.

Asit Srivastava: That's good. Who joins besides Sarah? Which anybody else from the management team joins?

Vaibhav Singh: Yeah.

Asit Srivastava: Does Arun join or Arun's boss Todd or anybody? Do they join these

00:17:56

Vaibhav Singh: So I know four people that are higher level.

Asit Srivastava: calls?

Vaibhav Singh: S apart from S there's uh Arun joins not on a regular basis but every other day I'm seeing Arun. Deepak also joins.

Asit Srivastava: Okay.

Vaibhav Singh: Todd also joins uh Deepak what's say Aru Mogan. Aru Mogan.

Asit Srivastava: Okay.

Vaibhav Singh: Okay.

Asit Srivastava: Okay.

Vaibhav Singh: Yeah.

Asit Srivastava: Deep is is an SVP is like four levels, five levels above.

Vaibhav Singh: Yeah. Deepak Kumar.

Asit Srivastava: Uh okay.

Vaibhav Singh: Aru Mogum. Aru.

Asit Srivastava: All right.

Vaibhav Singh: Yeah.

Asit Srivastava: I know that. Yeah. Okay.

Vaibhav Singh: H he joins sometimes but he joins as well and then Todd

Asit Srivastava: That's okay.

Vaibhav Singh: joins once in a while and there's another guy I don't have any interaction with

Asit Srivastava: Okay.

Vaibhav Singh: him but it's Elar his name Eldar Cassid something he joins

Asit Srivastava: Okay. Got

Vaibhav Singh: again he will join rarely on Mondays or sometime in between if he has something to say or some or show or

00:18:45

Asit Srivastava: it.

Vaibhav Singh: present in the Tableau dashboards he will show and I think he's the project lead or I don't know this exact title but he's there he joins so these four

Asit Srivastava: probably. So,

Vaibhav Singh: people

Asit Srivastava: uh there's only one call daily with the Sah, right?

Vaibhav Singh: yeah but this standard call is there with just

Asit Srivastava: Okay. Okay.

Vaibhav Singh: S

Asit Srivastava: I have one call on my uh I think Samir sent me the the series. I have one call on my calendar. It's a I guess it's a daily. It happens.

Vaibhav Singh: yeah that's

Asit Srivastava: But let's see what Okay. All right.

Vaibhav Singh: Okay.

Asit Srivastava: All right. I'll I'll let you guys go on. Okay. Um,

Vaibhav Singh: Okay. Okay.

Asit Srivastava: thanks me for the call.

Vaibhav Singh: Thank you. Thank you. Thank you. You can drop as well if you want. It's okay.

Saivijaay VK: Bye bye. Bye bye. Take care, guys.

00:19:49

Saivijaay VK: Bye.

Vaibhav Singh: Okay. Uh Avin uh what I'll do is Samir. Okay. You had a chat with Samir. Hey Samir, you joined.

Sameer Shaik: Uh yeah.

Vaibhav Singh: Yeah.

Sameer Shaik: Uh like he didn't

Vaibhav Singh: Great. Uh I have set a meeting at 7:30 like uh that will be half an hour from

Sameer Shaik: get

Vaibhav Singh: now.

Avinash Patel: Yes.

Vaibhav Singh: So are you okay? Oh yeah. So Sami you had a blocker in that time.

Avinash Patel: Yes.

Vaibhav Singh: Should I is there something else or what's what? What what what is it? I just set it up. I don't know.

Sameer Shaik: blocker.

Vaibhav Singh: Yeah. And it was saying you had some conflict at that time. Half an hour after this. So anyways,

Sameer Shaik: Oh, I don't know.

Vaibhav Singh: I'm just saying there's should after our daily stand up, we'll have a stand up of our own. Okay, that's it.

Sameer Shaik: Okay, that is that is a the different project that is for only today I

00:20:47

Vaibhav Singh: Yeah. Oh okay fine.

Sameer Shaik: guess.

Vaibhav Singh: Uh okay so what I'll do is I'll give the stage to

Avinash Patel: Oops.

Vaibhav Singh: you. Uh just explain to Abinina so Ainaash is also knows what's going on with the project. So you are working on the MLOps side. Just give a brief overview of what has been done what's going to happen any challenges anything. Okay. Fine. Okay.

Sameer Shaik: Yeah.

Vaibhav Singh: Hurry.

Sameer Shaik: Okay. So, uh currently uh hi first thing. Hi. So, I think like we met I've seen you but we have to meet in the office like probably tomorrow or

Avinash Patel: Yeah.

Sameer Shaik: Monday.

Avinash Patel: Yeah.

Sameer Shaik: So currently what I'm working on is like we have building this MLOps pipeline. So in the on GCP so the plan is like currently started with the YOLO X model single model object detection model. So so we upload the data in the Google data bucket. So once the data is uploaded there then entire training and logging and monitoring everything should happen automatically.

00:22:00

Sameer Shaik: So for now this is what we planned. So what I what I've started doing is uh using the notebook. So I've training the models model on the notebooks like uh allocating the instances GPU instance and all the resources. So I started with that. So it was successful few days ago and now what I'm doing like since like we we have to use this custom models right. tool X and all like we this a custom training script so we can't use the AutoML and all so so it was successful the basic training like we just pull the data uh get the repo from the GitHub and we're pulling the repo then adding some uh using the images that are there in the Google bucket so that training automatically triggers in the using the Vertx AI pipelines so so we have to use the entire Vortex AI ecosystem Google Vortex ecosystem. So where they have all the experiment tracking

and monitoring and also data data data sets and also we can also integrate like uh deployment as well endpoint at the end.

00:23:07

Sameer Shaik: So currently so the training is happening and logging and monitoring is also done. So I worked on that and also the data sets. So data sets what it does? So we can visualize the data set like what are the number of classes over there because in the Google data buckets we can't visualize the data. So we can see but we can't visualize. So that is advantage of data data set. So that I've integrated that and also I've integrated the vortex a experiments vortex a experiments just for monitoring and logging all the hyperparameters and artifacts and all. So that is also done. And then previously I started working like using the GitHub repository and then using the that repository for training. So now I have I'm using this docker. So I'm first I'm building the docker docker container then using the cloud build. So after that like I'm using the docker image for the training purpose. So that is also done. So these things are done as of now.

00:24:06

Sameer Shaik: So and also validation. So once the training is done, it evaluates like all the model uh checkpoints and everything were logged back to the Google data buckets itself. So everything the data the storage everything lies in the data bucket and remaining things will be like uh monitoring and experiments and this we for visualization purposes. So as of now these things are done evaluation and all these things are done. Oh

Vaibhav Singh: Just a question.

Sameer Shaik: yes,

Vaibhav Singh: So that retraining re-triggering part is going on or is it done retrigger? So I have a model I built a model I trained a model I have some metrics about

Sameer Shaik: re-triggering.

Vaibhav Singh: it and the model is in production let's say right and after a time I start seeing somehow somewhere the model performance is degrading

Sameer Shaik: Auto trigger is not done like automatically trigger.

Vaibhav Singh: okay

Sameer Shaik: So that is to be done and also like we need like how we have to implement like if the

Vaibhav Singh: Yeah.

Sameer Shaik: data like if the if the data is same like we need to re-trigger this pipeline every day or something like using

00:25:22

Vaibhav Singh: Yeah.

Sameer Shaik: the same data or the data changes. So we need inputs on that like this is the phase one right.

Vaibhav Singh: Yeah.

Sameer Shaik: So now what I'm trying to do I just have these metrics and all the details for training purpose. So we can next step like we can integrate we can create a uh docker image and then push it into the registry then we can deploy automate the deployment as well as a next step and once that is done we can integrate this

Vaibhav Singh: Okay.

Sameer Shaik: event trigger using the popsup or event RC. So this one like what it does it automatically triggers this training whenever this event happens. So whenever there is a change in the Google data bucket or something.

Vaibhav Singh: Okay.

Sameer Shaik: So based upon like what we discuss and

Vaibhav Singh: Yes. So the current deployment you know the current deployment method it's like

Sameer Shaik: repair.

Vaibhav Singh: repository of this repository of that and there's action Google GitHub action that takes care of things right so you going to integrate that itself or what's the and deployment in non-production right okay

00:26:19

Sameer Shaik: Yes.

Vaibhav Singh: so and

Sameer Shaik: currently like yeah like whatever the method that we are doing to deploy right so the end points to the vertex endpoints the same thing like I'm planning to integrate that till the end

Vaibhav Singh: Oh okay.

Sameer Shaik: points yeah no yeah no yeah no yeah no yeah no yeah no yeah no yeah no yeah no yeah

Vaibhav Singh: Oh yeah. Yeah. Now I understand. So this is just model stuff.

Sameer Shaik: no processor is different

Vaibhav Singh: Processor is not different. Okay. Yeah. Correct. Correct.

Sameer Shaik: yeah so

Vaibhav Singh: I combined that both of them. No worries. Sounds good.

Sameer Shaik: this tech I'm going to try on the home depot uh vdi because like I'm not sure like whether we'll be having all the accesses and all and also we need to integrate the service account because currently I'm using this direct account thing. So I'll check with that.

Vaibhav Singh: Yeah,

Sameer Shaik: So once that is done then yeah I think like we can go with the next phases and integrate other things as well.

00:27:09

Vaibhav Singh: look.

Harish Kumar S: So this is for a single framework you are mentioning some like yolo.

Sameer Shaik: Yeah,

Harish Kumar S: Okay.

Sameer Shaik: actually like because this is these are custom right like if if other models are in

Harish Kumar S: Okay.

Sameer Shaik: the same format then we can use like it is flexible to extend but if it is a different framework or something then we have to integrate like

Harish Kumar S: Yes, because for each model we have a different GitHub proper, right? Oh yeah, that's it.

Vaibhav Singh: Okay. Uh let me know when you'd start doing those stuff.

Sameer Shaik: So

Vaibhav Singh: Uh the thing is anything that you are doing right now S has no idea about and I don't think Jared also knows anything about this things.

Sameer Shaik: yeah,

Vaibhav Singh: They're working on it in the background. I'm not sure.

Sameer Shaik: so we'll deploy in the home depot machine and then we'll create some

Vaibhav Singh: So the the stance yeah the stance right now over there as far as I understand is you and Fatima were doing labeling.

00:28:09

Vaibhav Singh: Okay. And Harisha as well and me as well and Salanjit who is not in the project anymore is sick but that's not the guy who's sick.

Sameer Shaik: Hello. We lost you, I guess.

Avinash Patel: Oh,

Sameer Shaik: I think he's out of network or like something

Avinash Patel: speech speech.

Sameer Shaik: even Harish was frozen. Oh, I think they both are in the same office.

Avinash Patel: Uh,

Sameer Shaik: So something might be similar happened but both of

Fatemeh Teimury: Yeah,

Sameer Shaik: them

Fatemeh Teimury: I think

Avinash Patel: it's

Fatemeh Teimury: so.

Avinash Patel: fun.

Sameer Shaik: Okay.

Vaibhav Singh: gone.

Sameer Shaik: Yeah.

Vaibhav Singh: Yeah. So, uh Okay.

Sameer Shaik: I'll start web today.

Vaibhav Singh: Probably.

Sameer Shaik: So what we'll do I'll like I'll integrate I'll try to deploy in this home depot

Vaibhav Singh: Yeah.

Sameer Shaik: machine.

Vaibhav Singh: Mhm.

Sameer Shaik: So when and also create some basic presentation like what's happening and what are the planning things that we are going to

Vaibhav Singh: Sure.

Sameer Shaik: do in the next steps maybe and then we can present it sometime like so that like some work will be going on

00:29:31

Vaibhav Singh: Yeah. Yeah. Sure. Sounds good.

Sameer Shaik: that

Vaibhav Singh: This ML of stuff is kind of important. Uh you also mentioned it's kind of something we have to do this. That's not optional. The single line that we are working on in is fine. We are working on it. ML of stuff we have to build.

Sameer Shaik: Yeah. And also Weber do you have when I asked Saranjit like he mentioned like we pitched this MLOps pipeline

Vaibhav Singh: Okay.

Sameer Shaik: but so do do you have any story or something that we have to do or any there any requirements mandatory requirements from them because like the requirements are currently not nothing it's vague.

Vaibhav Singh: What I know is we okay crude form is this. You don't have a millos pipeline. This is how you do MLOps. That's it. Okay. So that that is what we told them and that is you I'm not sure if you acha you guys were

Sameer Shaik: Oh,

00:30:27

Vaibhav Singh: not there at the time or maybe so we had created an entire flowchart diagram and explaining okay this is supposed to happen this maybe Sanjit has shared an image with it's a big blue color diagram

Sameer Shaik: that reflect everything with this.

Vaibhav Singh: something yeah very too many boxes so

Sameer Shaik: What?

Vaibhav Singh: everything is something we just said okay you don't have this you don't have this this is how we what we'll

Sameer Shaik: Yeah,

Vaibhav Singh: do that was told that I was told 3 months ago right now I don't think the

Sameer Shaik: I think yeah,

Vaibhav Singh: videos even started properly okay fine no worries it's fine uh

Sameer Shaik: I think this is helpful maybe like yeah automating deployment and everything training and

Vaibhav Singh: yeah

Sameer Shaik: all but yeah so for all the models if you have to implement all the models then a lot of work have to be

Vaibhav Singh: so just be aware of this there will be a lot of push back the guys have been deploying stuff for four years now they are used to a certain certain way of they work used to

00:31:23

Fatemeh Teimury: I need Guys,

Vaibhav Singh: work a certain way of working there will be a lot of push back there will be some egos that will get hurt so the best way forward is to have all the questions beforehand go to chat JP and ask okay how to not hurt the ego okay get all the questions that you can conjure up and just be ready for the questions because they will He asked so if you guys remember in

Sameer Shaik: Nice.

Vaibhav Singh: the Deep meeting itself on what was it Monday or some other time what what did he say he said I don't think it will work please test it that's that's all the meeting uh 1 hour meeting okay so just be careful whatever you're doing okay okay is there any

Sameer Shaik: Sure

Vaibhav Singh: other work that Satish and the other guy has to help you with or what's it their involvement is done or they will be

Sameer Shaik: actually like Satish is just like he just gave the access and all like sir like he

Vaibhav Singh: involved.

Sameer Shaik: like asked couple of two two to three times like he's also busy with some other projects and all so I'm like exploring and doing alone like as of now like that's nothing much because they even

00:32:40

Vaibhav Singh: Okay. Okay.

Sameer Shaik: like didn't commit like he didn't commit like he'll not deploy he'll just help like whenever there is some time that's it so can't rely on

Vaibhav Singh: Oh okay.

Sameer Shaik: them

Vaibhav Singh: Just let me know if you have any requirements or doubt. I probably won't be able to solve anything related to JCP but at least idea discussion we can all have.

Sameer Shaik: yeah like it's it's bit clear for now everything like using this even like we

Vaibhav Singh: Okay.

Sameer Shaik: got cursor and like charg and now I'm getting more.

Vaibhav Singh: Let's have you guys all connected to cursor right? Yes.

Fatemeh Teimury: Yeah.

Vaibhav Singh: cursor.

Avinash Patel: Yeah,

Vaibhav Singh: Okay,

Avinash Patel: just got

Vaibhav Singh: I have to do I have to do that.

Sameer Shaik: Yes.

Vaibhav Singh: Uh,

Avinash Patel: yesterday.

Vaibhav Singh: one thing that I want Fatima and Samir to take care of is Samir. It's Samir. Ainina might not. I mean, Ain, do you have the VPN access?

Avinash Patel: No.

Vaibhav Singh: No. Okay.

00:33:38

Vaibhav Singh: So, the GPU and the VPN access. Just help him with that. Okay. Please.

Sameer Shaik: So,

Vaibhav Singh: Okay.

Sameer Shaik: so we'll be getting the accesses and files whatever I

Fatemeh Teimury: Oh, just just wanted like to mention something.

Vaibhav Singh: Yeah.

Sameer Shaik: need.

Fatemeh Teimury: I still don't have access to 108 since yesterday. I actually like give me some time.

Vaibhav Singh: Oh, okay. What?

Fatemeh Teimury: I last night on maintenance.

Vaibhav Singh: 10.

Fatemeh Teimury: So I'm just checking if you guys

Vaibhav Singh: Let me uh s***. Let Let me just What's the IP address?

Fatemeh Teimury: like

Vaibhav Singh: Full IP address? Uh 17216.

Fatemeh Teimury: this

Vaibhav Singh: Sorry

Sameer Shaik: It's 9:30.

Fatemeh Teimury: 20108.

Sameer Shaik: I think we have to join that meeting.

Vaibhav Singh: there's no meeting today. Thursday.

Sameer Shaik: Oh, okay. Nice.

Vaibhav Singh: It's Thursday. Uhhuh. Ain,

Fatemeh Teimury: Yeah.

Vaibhav Singh: one other thing.

Sameer Shaik: Good.

Vaibhav Singh: Thursday is no meeting. That's good.

00:34:31

Vaibhav Singh: Otherwise, every day daily there's this time there's a meeting. There's a standup daily standup.

Sameer Shaik: Thank

Vaibhav Singh: Uh Fatima Kishor is in office today.

Avinash Patel: Okay.

Vaibhav Singh: Just irritate him for this please.

Fatemeh Teimury: Okay. Yeah, sure.

Vaibhav Singh: Okay.

Fatemeh Teimury: I did couple times.

Vaibhav Singh: Uh okay. Bug bug him till it doesn't work.

Sameer Shaik: you.

Fatemeh Teimury: Okay.

Sameer Shaik: Even I don't have access to that.

Fatemeh Teimury: Can you can you come in? Oh, okay. Yeah. So,

Vaibhav Singh: 108.

Fatemeh Teimury: pick up pick up last night.

Sameer Shaik: Maybe

Vaibhav Singh: You guys don't have You guys had access to 108 last I

Fatemeh Teimury: Yeah.

Vaibhav Singh: remember.

Fatemeh Teimury: until last until yesterday.

Vaibhav Singh: Okay.

Fatemeh Teimury: But like basically Oh, t-shirt is fixing me.

Vaibhav Singh: Okay.

Fatemeh Teimury: Okay,

Vaibhav Singh: No worries.

Fatemeh Teimury: I'm going to check that.

Vaibhav Singh: Okay.

Fatemeh Teimury: I'm going to like update you guys. Um one way connection,

Vaibhav Singh: Yeah.

Fatemeh Teimury: but I was disconnected from VM side.

00:35:17

Vaibhav Singh: Okay.

Fatemeh Teimury: It was not like my Yeah.

Vaibhav Singh: I think that I think Shamir you are done with your site right what work you're doing you're working on

Sameer Shaik: there.

Vaibhav Singh: the envelope site okay and what's going on next is done Fatima can you go next and if you want you can share your screen and show you the results and I'll also get some update

Fatemeh Teimury: Um yeah, I think just like kind of like explain what is going on on my side.

Vaibhav Singh: regarding

Fatemeh Teimury: So whatever S mentioned, I'm working on the model the side. I'm helping with that. So basically like uh we're trying with um detection and segmentation models yolo based so far

Vaibhav Singh: probably show an image.

Fatemeh Teimury: to

Vaibhav Singh: So probably better to show a ground image.

Fatemeh Teimury: think I'm having okay yeah

Vaibhav Singh: So aas to give a little context uh the visual embeddings pipeline that we want to build for them has two parts. One of them is in uh model based inference that is we want to detect the

00:36:13

Fatemeh Teimury: Nice.

Vaibhav Singh: products what products are there in the selling area right so right

Avinash Patel: Yep.

Vaibhav Singh: now they are in every model deployment they have they don't detect the products they only detect the price labels and the empty regions they don't detect products so to do product embeddings visual embeddings product has to be detected So what we what Fatima is

Fatemeh Teimury: Yeah. So I kind of like

Vaibhav Singh: working on is the first part of the pipeline we'll pass the image to a model detection segmentation or object detection we'll get the products then after this crops of each product has been extracted it will go to another pipeline for the visual embedding. So over there the entire embedding related stuff will happen. We'll apply dynino v3 sig sinclair whatever works whichever works in conjunction whichever works create a history store the embeddings uh do some retrieval task and help with the sk matching. So that's the python. So this is first pip first part of the pipeline. Okay.

Fatemeh Teimury: Yeah. So I can like um add a little bit to Yeah.

00:37:30

Vaibhav Singh: This is a really bad

Fatemeh Teimury: I mean I because I'm not connected I don't have like other annotations.

Vaibhav Singh: emotion.

Fatemeh Teimury: Sorry about that. Well, like basically like let me give you like a little bit of um context on the data like add add on top of like what mentioned. So basically like we have two different data and we're like having a bunch of um bounding boxes uh with different like classes. So so far we have like product price tag shelf beam and empty. Initially we started like focusing on taking empty boxes because the whole idea is to like come up with a um like supply chain to take off like the empty uh part of the shelves and like uh optimize that. But uh apparently on the Home Depot side, they're using like separate models for each classes and they have like good results with for like detecting price tags, shelf beans and uh currently we're focusing on

Vaibhav Singh: What is

Fatemeh Teimury: detecting product and cluster product.

Vaibhav Singh: that?

Fatemeh Teimury: So um for that uh we tried like some uh detection and segmentation yolo based models and uh there are like a some like some problems to basically like um increase the performance.

00:39:00

Fatemeh Teimury: One is uh regarding the data annotation because um product and because basically like Cer products are like just like um a combination of like few products and then um for YOLO based they're doing like pixel level like detection and the model kind of like confused that. So what I'm doing I'm uh trying like to come up with um uh some sort of like uh method to improve the product detection or segmentation uh and then like we can uh basically like um you deploy to Home Depot as like a single model only focusing on product and plus their product. So what I've been uh what what I've tried like so far I tried like s in but apparently like um it's not like it

Vaibhav Singh: f*****

Fatemeh Teimury: was not like adding enough to the uh pipeline. The other thing that uh I'm thinking is like basically like initially

Vaibhav Singh: heat.

Fatemeh Teimury: have one uh stage of like detecting product. So considering both cluster product and product as like one class and then based on the uh the distance between products then we can cluster them.

00:40:31

Fatemeh Teimury: So maybe like we can we can improve that and I've also like the other day suggested like to change the uh the threshold for detection. So I've also like did that it added like a little bit to the pipeline but not that much. So the main like the big picture of what I'm working on is to come up with different vision based like uh pipeline that is powerful for detecting product and cluster product uh based on the data that we have and the data as you can see. So right now we have like what I'm working on is just like single frame um photos of the store containing like uh different types of product um and

it's pretty dense. the resolution is down like four 5K and uh yeah so we tried like basically like I've also like tried uh to um like uh just how to say that like just like um train it on for example like 1K like um 16 images just like of 1K okay for a single image but that that didn't work

Vaibhav Singh: Oh,

00:41:58

Fatemeh Teimury: like it's just like the data is dense we have like so many classes in

Vaibhav Singh: okay.

Fatemeh Teimury: like single image and then the nature of the problem is that is detecting like pixel base for product and cluster product and the way that the data was annotated initially by the team they are not They didn't follow the same approach for data preparation.

Vaibhav Singh: So, Fatima just looking at

Fatemeh Teimury: Yeah.

Vaibhav Singh: this image itself, there's a lot of products for a single price tag,

Fatemeh Teimury: Mhm.

Vaibhav Singh: Right.

Fatemeh Teimury: Yes.

Vaibhav Singh: And there is so they don't have a cluster product around them.

Fatemeh Teimury: Yeah.

Vaibhav Singh: They don't Okay.

Fatemeh Teimury: Mhm.

Vaibhav Singh: I feel like this is just a thinking. Maybe it's will be maybe in regions if we get them to reannotate the images. Okay.

Fatemeh Teimury: Yeah, that would be very helpful.

Vaibhav Singh: For cluster products and products. So in images what's similar to what sashi asked them to do.

Fatemeh Teimury: Yeah.

Vaibhav Singh: So create a cluster right.

Fatemeh Teimury: Yeah.

00:43:18

Vaibhav Singh: So instead of in these images if a cluster product is available we can skip the inside products and maybe we can do some prediction separately on that particular region.

Fatemeh Teimury: So you mean like for the first stage I totally ignore the spot I

Vaibhav Singh: So yeah. Yeah.

Fatemeh Teimury: provide?

Vaibhav Singh: Yeah. So let's let's uh let's take the uh right

Fatemeh Teimury: Oh, okay.

Vaibhav Singh: column sorry left column second row actually the second column the

Fatemeh Teimury: This one. F1.

Vaibhav Singh: last image then below this uh this can be we also yeah same product

Fatemeh Teimury: That one.

Vaibhav Singh: multiple uh product right multiple annotations if just in case there was a cluster product around that this entire thing something like And I only focus during training focus on training on detecting the products cluster product instead of the product and when I am sure it's a cluster product I have model as detected it's a cluster product let me work separately on that particular patch of image to detect individual products because there

Fatemeh Teimury: H I see.

00:44:29

Fatemeh Teimury: So she cluster product and then we check product inside the cluster product.

Vaibhav Singh: there's think of yeah think of it like

Fatemeh Teimury: Okay.

Vaibhav Singh: this if I have 10 embeddings,

Fatemeh Teimury: Okay.

Vaibhav Singh: 10 images of the same product. It's not going to help me that much, right? So, this has

Fatemeh Teimury: Because I was like thinking the other way that makes sense like um this

Vaibhav Singh: to.

Fatemeh Teimury: like to a stage network like also came up to my mind. So I was thinking like initially for the data I assume product and cluster product as a single class. So first like detect everything right and

Vaibhav Singh: No, no, no. Okay.

Fatemeh Teimury: then and then let's say let's say like we have some product embedding for example you see like this you see my mouse right well these products they look same and their

Vaibhav Singh: Yeah. Yeah.

Fatemeh Teimury: embedding should be similar almost then I can cler

Vaibhav Singh: Exactly. Yeah.

Fatemeh Teimury: them I think

Vaibhav Singh: So that is what my thinking a little bit different from what you're approaching but

00:45:26

Fatemeh Teimury: it was like different first detect and then take

Vaibhav Singh: but yeah so it doesn't really matter as long as I can detect the product properly and that belongs to the same price tag right if I if there

Fatemeh Teimury: products.

Vaibhav Singh: are 10 products I don't detect 10. I only detect one. It's fine. That is my

Fatemeh Teimury: So,

Vaibhav Singh: thinking.

Fatemeh Teimury: okay, just just as I just wanted like to be clear on that we are on the same page. Let's say like here, right?

Vaibhav Singh: Yeah.

Fatemeh Teimury: I have like I don't know five, six, seven something like similar.

Vaibhav Singh: H.

Fatemeh Teimury: So what you're saying that if I detect two products then I'm

Vaibhav Singh: Yeah.

Fatemeh Teimury: fine

Vaibhav Singh: I'm saying one is also fine as long as the one is detected properly.

Fatemeh Teimury: for one small product.

Vaibhav Singh: one product per price tag is good enough one good

Fatemeh Teimury: Huh? Okay. Okay. And then what should I

Vaibhav Singh: visual I'm thinking from the point of the

00:46:15

Fatemeh Teimury: do?

Vaibhav Singh: pipeline I understand when we are building the second pipeline as many embeddings that we can get it's will be a good thing but this has to go as an input to the second pipeline right and

Fatemeh Teimury: Um

Vaibhav Singh: over there I'll get embeddings the almost the same embeddings via any model or Any approach I apply the vector embedding information will be the

Fatemeh Teimury: I'll see so basically like they say like detecting products has

Vaibhav Singh: same.

Fatemeh Teimury: higher priority than taking cost of products for embedding.

Vaibhav Singh: Yes. Yes. Almost.

Fatemeh Teimury: Okay.

Vaibhav Singh: I would Yeah,

Fatemeh Teimury: Okay.

Vaibhav Singh: you can say that.

Fatemeh Teimury: Okay.

Vaibhav Singh: But yeah.

Fatemeh Teimury: Cool. Good to know.

Vaibhav Singh: Okay. Yeah. But I I'm not sure how this will be achieved.

Fatemeh Teimury: Yeah.

Vaibhav Singh: that might be read might need some reannotation of the data itself of I don't

Fatemeh Teimury: Honestly,

Vaibhav Singh: know now I'm going to ask you something again please

Fatemeh Teimury: like

Vaibhav Singh: get connect with snare sheish or harish get the data pending data that we have we have about 9,000 images yes that has never

00:47:24

Fatemeh Teimury: Oh, really? I thought I sit. I thought I need to get it from Oh,

Vaibhav Singh: no no yes get

Fatemeh Teimury: so and Harish, right? And this is like the how many how many characters they have in the new

Vaibhav Singh: the data Please there is no data that's what I'm going to

Fatemeh Teimury: data?

Vaibhav Singh: say we have to pre-annotate them to send it to Sum Sumit so he can annotate

Fatemeh Teimury: Oh, that would be great.

Vaibhav Singh: it okay please please please do

Fatemeh Teimury: Yeah, I'm going to I'm going to Yeah. Thank you so much.

Vaibhav Singh: it okay so Ainas do you have any questions about this one so

Fatemeh Teimury: Okay. Yeah.

Vaibhav Singh: I know it's was a little too dense

Avinash Patel: Yeah,

Fatemeh Teimury: Yeah.

Avinash Patel: I I'm just going to probably look through uh all the sort Can this document be like sh is it shared with me or can it be

Fatemeh Teimury: Yeah.

Avinash Patel: shared with me or Yeah.

Fatemeh Teimury: I'm gonna I'm gonna send it to you.

00:48:06

Avinash Patel: So,

Fatemeh Teimury: Sure.

Avinash Patel: I'll just read it over and then any sort of like software exploration I'll just do by myself and then I'll get used to

Fatemeh Teimury: Okay. Yeah. So, that would be good.

Avinash Patel: everything.

Fatemeh Teimury: Yeah. And let me know what

Vaibhav Singh: So ain I also want you to think of a problem. So think about this. In a store where there's supposed to be products, empty is an anomaly, right? It's empty is a concept. It's empty reg. There are many empty regions. There's it's it's why is it empty is only based on if there's a price tag available or not. That is how we are determining if there's a price tag associating and only then it's empty. So can MT be treated as an anomaly and can I run do something with anomaly detection anomaly clip? There are two three papers of anomaly clip. There are multiple anomaly detection

Fatemeh Teimury: Hello. Sorry, I'm just like interrupting you here.

00:48:58

Vaibhav Singh: models.

Fatemeh Teimury: Just mentioning that like the pipeline that we have so far is detecting price like pretty well.

Vaibhav Singh: Okay.

Fatemeh Teimury: So maybe we can use it.

Vaibhav Singh: Yeah, exactly.

Fatemeh Teimury: Yeah.

Vaibhav Singh: So I don't care if the problem gets broken down as gets solved by two three different models. I care the problem is solved. If for this model product and price tag is working really great and empty is not working fine, fine, no worries.

Fatemeh Teimury: Nothing.

Vaibhav Singh: The home depot already has something for empty. But if we are want to work on the empty detection and if you want to approach it via some other way that's fine but it has to work. Okay. Yes

Fatemeh Teimury: Yeah.

Vaibhav Singh: sir.

Sameer Shaik: So what is cluster product here? Can you explain the Fatima?

Vaibhav Singh: Cluster product actually the idea of cluster product started when we first started the project is to help the annotation team to annotate faster. So every week we had to uh me and Harish had to present something present improvements every week.

00:50:15

Vaibhav Singh: So we started with 600 images and the cluster product idea. If you look at the bottom left corner image we asked them to annotate instead of annotating single products just annotate the entire product as cluster product. And the idea was we'll later come back to it and then when we have enough time enough bandwidth then we'll annotate single products. But if you just look at the image on the right sometimes they have already annotated everything as single single products they have not marked. So there's a little bit of inconsistent the meaning of cluster product and when it has to be done. This is one cluster product. There's another cluster that she wanted for visual embedding. So that means when there are two products side by side that looks the same has the same price tag that he can use for ext for his embedding extraction

because right now if there's 20 products in a bay each product will be different and right he'll just have 20 products but he'll just have one crop per product that is not enough right so his idea was so and there are cases where multiple products, same products are there on the same shelf, same hanging area.

00:51:33

Vaibhav Singh: It's the same product, but there are multiple rows of it, multiple items. They also have their separate price tags as well. It's the same, but it's the same product. Think of it like this. So, there's a mass uh real fruit juice. It's there's kept in one one row of the shelf, it's also in the another row of the shelf. So the idea is to the what he wanted the annotation team to do is just

Avinash Patel: Mhm.

Vaibhav Singh: mark those together just add a clust one more bounding marks around them apart from the individual product item labeling so that's another there's a little terminology overlap but it's okay fine for now

Sameer Shaik: Yeah, that makes sense.

Vaibhav Singh: so like what I was saying last is please think of how to work on the improvement of the product. Price labels I understand price label is high that is expected products is something entirely new. We need all the brain power that we can apply to solve the product thing. Okay. Right.

00:52:43

Fatemeh Teimury: Honestly,

Vaibhav Singh: So this is one of the problems.

Fatemeh Teimury: I feel like I'm going to give it a try after our call, but like I feel like if I completely drop cluster products, Then like let's say like the model already see like this product,

Vaibhav Singh: H

Fatemeh Teimury: right? I think they're going to detect product here. So then we can get like product and then based on their

Vaibhav Singh: what I what I used to do was merge cluster product and product doesn't

Fatemeh Teimury: embedding.

Vaibhav Singh: matter.

Fatemeh Teimury: Okay. Okay. Okay. Okay. Cool. Yeah, that that also works. I'm going to give it a try. Okay,

Vaibhav Singh: Yes,

Fatemeh Teimury: sounds good.

Vaibhav Singh: just give it a try. Okay,

Fatemeh Teimury: Feel

Vaibhav Singh: so problem number one is anything MLOps related stuff Samir is working on,

Fatemeh Teimury: like

Vaibhav Singh: he might have some doubts. You have experience with GCP, right, Ain?

Avinash Patel: Uh yeah, getting some more recently, especially since with um with the other Home Depot project, we're integrating a lot of GCP,

00:53:41

Vaibhav Singh: Yeah.

Avinash Patel: so getting pretty exposed to

Vaibhav Singh: Okay.

Avinash Patel: it.

Vaibhav Singh: So problem one helping Samir and anything GCP related or the MLOS pipeline related. Problem two working on how to improve the product detection pipelines with Fatima. Problem three is something I guess Harish will explain to you now. Okay. So Harish is working on the single line stuff and there are two problems in the single line that I know of. There may be many but probably let him present something and then I'll come to the problem.

Avinash Patel: Okay,

Vaibhav Singh: Okay.

Avinash Patel: sounds good.

Harish Kumar S: Hey, good morning everyone and first of all welcome to the team a webinas.

Avinash Patel: Thank you.

Harish Kumar S: So, so for like uh yeah, so we are working on this uh single line detection problem and uh as you can see like so the FIMA show some product area right so in WEO they divide the bay into two segments like it's a selling area and it's an overhead area okay so selling area is mainly where the people purchase their products by hands.

00:55:04

Harish Kumar S: So it's like they get the products directly from the bay and override area is like a stock of the particular uh respective selling area. So they maintain the stocks of the all the products that belongs to the respective base. Okay. So I will share my screen. So my computer is also getting needs to get rebooted now. So, I will start that. Give me a second.

Vaibhav Singh: Harris is facing technical problems. What happened?

Harish Kumar S: A patch reboot.

Vaibhav Singh: Okay,

Harish Kumar S: Five minutes.

Vaibhav Singh: so there are technical issues over here. Uh so remember we were I was talking to you about yesterday the overhead region. So it's the same thing. So there's single line SQS over there and there's multi-line SQS over there. Now problem with the single line SQ is we have solved the pipeline so far with object detection segmentation and Google OCR. Like I said yesterday, Google OCR is sensitive and the image size the stacking method that we applied has benefited us but I was not expecting it to work.

00:56:28

Vaibhav Singh: It has given us boost. It makes no sense why there's a boost. So another problem with Google OCR is uh the latency as well as there's another cost. Now their team for the another pipeline for multiline they have already shifted to their own custom digit detection model. Okay. And we want to do something similar. Now the problem, one problem is the SKU labels that Home Depot team was working with for multi-line had some anchor points on the image that they can use to rotate the images. So let's say there was some indicator in the label itself whether the label is tilted or it's upside down there was some indicator but in the single line SQ labels that we are working with the classes that we have don't have any indications whether it's upside down there are cases where you we can say it's up up or down but it's very rare and majority of the labels won't you won't see any indication one job replace the Google OCR with a better OCR or custom OCR or custom dete digit detection model but problem with digit detection model is we need to do something with about the orientation maybe the OCR offtheshelf OCR may be able to work as is don't know maybe the digit detection algorithm that they have already worked on may work.

00:58:08

Vaibhav Singh: I also don't know about that. So problem one and two is creating finding out a method or way. Okay, sorry. The first is we have to find out how many SKU labels are oriented and how much orientation the either the Google OCR or existing OCR that another OCR of the shelf like GLM OCR or DPC OCR or nanonets OCR will use is create is not able to detect that's one the second is if the number is too high how to do the orientation correction Third, let's replace the OCR. That's far problem, far in the future problem, but it is one of the things. Okay. So, working.

Harish Kumar S: No.

Vaibhav Singh: Okay.

Harish Kumar S: Okay. Okay. I think you can see my screen.

Avinash Patel: Yes.

Harish Kumar S: Yes. Okay. So, as I said before like uh so this is a entire bay. So you can see like the this is a bay number. So the below the segment below this are the selling area.

00:59:35

Harish Kumar S: The selling area where we have the price labels so that the people can take up the things and purchase it and above this self is the stock area. So they mention it as a overhead area. So this is where the stock of the particular uh respective bay items are presented there. So when the uh when the product when the selling products are empty so they will restock it using the those products. Okay. So for each in the overhead area there for the each stock they have the respective skew labels like this. So as VBO mentioned so we have multiple types of uh skew labels. So this they call it as a pallet and uh one second. Okay. So this they call it as a RDC and this is the multi-line. So multi-line they have created a detection model by the home report and uh and they get a skew from that. So for that we have created so initially what they have done is so they did a multi-line and uh they use the Google OCR to get the skew values but they they are not getting the accurate label.

01:01:02

Harish Kumar S: So what we have done is they asked for the segmentation model over the multi- lines. So once they get a crop of all the multi-line label they create we create a strip and we do the segmentation on it to classify it to segment it as a first line second line and the quantity. Okay. So this quantity is like so they consider it as an anchor point. So that if the label is like a rotator they use this reference they use this as uh this quantity as an anchor point and determine which on which angle is got rotated. So they will correct an angle so that it will be u make it readable. So what we are doing is uh for the single line their existing pipeline is like uh so they send the entire image to the Google OR and the Google OCR gives all the extractions irrespective of the labels and everything and using the rule based and regrets patterns they filter it out and they get a skew lab uh skew values.

01:02:12

Harish Kumar S: So this is the existing pipeline. So what we have done is uh so we modify so we modify the pipeline into two stage. So first stage is like first we have to detect those single line labels from the overhead irrespective of the B. So I will show a sample image over there. Okay. So what we have done here is so we used detection as well as the segmentation. So for the detection we detect first we deduct uh can nice image. Okay let's say this is an over image. So first what we have to do is first we deduct all the single line levels. Okay. So we have five classes of uh five different types of single labels single line labels. So first what we have do is we will deduce all the single line tables. Then we stack it into a horizontal uh we stack all these crops into an horizontal strip and send it to a segmentation. Okay.

01:03:49

Harish Kumar S: Over the segmentation we what we have do we segment the respective numbers. You got my point? So uh after segmenting we what we have do we map those polygon points to this original image and we white out we white out the other noises. So and we send this entire mast d image to the google here to get a skew values. So this is how our pipeline is now working on and what we have observed is like uh we have got a boost of uh 5%age over there. So right now we are at point of 91.8%age 8 percentage and another part as VBO mentioned like the two things is like the OCR extraction so Google OCR uh we have to move on to more like u uh a task based OCR and other things is we have to deal with the rotated crops. Sometimes like Google was here is good at uh extracting the even at the label is like uh 90° oriented it can extract but if the label is like completely upside down uh it it does not extract anything.

01:05:17

Harish Kumar S: It extract but it's like always knit. So next thing we have to focuses on is the OR part for this pipeline. So anyone have any questions?

Sameer Shaik: Uh Harish uh one question like here we are getting multiple crops right so we are filtering out those all things this

Harish Kumar S: Yes. So using the regrets pattern we so the the final skew value must be either

Sameer Shaik: not

Harish Kumar S: six digits or 10 digits.

Sameer Shaik: so once after the OR we are filtering or before giving it to the O

Harish Kumar S: So we after the year uh we get annot like we get all the annotations text annotations and we use the regrets patterns over there to filter out false positives and other noises.

Sameer Shaik: Got it. Okay. I do.

Harish Kumar S: So this is how we will be getting the values.

Avinash Patel: Mhm.

Harish Kumar S: Okay, do you have any questions?

Avinash Patel: Uh, for now I I I should just uh explore whatever software you have in there before I'll I can ask questions.

01:07:11

Avinash Patel: Uh, do I get access to this uh codebase or is this kind of just your thing or do I get something else?

Harish Kumar S: as of now maybe I can share you this images share you those images so once

Avinash Patel: Okay.

Harish Kumar S: you get access to the machine you can so it's in a common data so we can display over

Avinash Patel: Okay, sounds good.

Harish Kumar S: there

Avinash Patel: Yeah.

Harish Kumar S: okay label.

Sameer Shaik: I think we are done moving back

Vaibhav Singh: over. Everyone is everyone knows who's what's working on who's working on what. Okay. So on me I'm not working on anything. I'm putting out fires wherever there's a fire. Okay. So that's my update. Tomorrow probably I'll start working again. All right. Okay. Uh it's okay. So what I've been working on, I've been working on data annotation for 3 weeks and then there's heavy labelings. I think my net is gone.

Sameer Shaik: Oh, we can hear you.

01:08:29

Sameer Shaik: We can hear you.

Vaibhav Singh: So whatever Google Meet is showing me you're trying to reconnect. Okay, no worries. uh probably tomorrow from starting tomorrow I have one thing in my mind for doing a single line verification job. So working on basically we have to find out where it's failing and how it's failing right Google OCR segmentation object detection we have to find out we know what metrics we are getting we are at 92 need to know where it's failing I have no idea where it's failing right now I need to know that so I have some script idea in my head I want to work on that I had started working on that 3 weeks ago but after that it's got dropped I'll probably start with it tonight or tomorrow morning And from tomorrow I think Harish is Harish and everyone is going to start working on the multiframe integration as well. I'm not sure what that work entirely entails but there is some work regarding that. Okay. Yeah, I think that's it. That's it.

01:09:29

Vaibhav Singh: It's too much for now. Too many too much information and I have no idea what Samir is working on what Fatima is working on. I've already forgotten one thing please. I have added everyone but I don't think anyone has filled their task list. Has anyone filled their task list?

Sameer Shaik: I'm I'm doing it.

Vaibhav Singh: So maybe says yes you're doing it

Harish Kumar S: No.

Vaibhav Singh: Fatima.

Fatemeh Teimury: Yeah, sorry.

Vaibhav Singh: Okay.

Fatemeh Teimury: I got to do

Vaibhav Singh: Okay. It's the end of the month.

Fatemeh Teimury: that.

Vaibhav Singh: There will be up some presentation I have to cook up for the this month end as well. Please fill it.

Fatemeh Teimury: Okay.

Vaibhav Singh: Please fill it in as much detail as you can.

Harish Kumar S: It's

Vaibhav Singh: Okay.

Fatemeh Teimury: Just Okay.

Vaibhav Singh: Okay.

Harish Kumar S: working.

Vaibhav Singh: Fine. Uh Ain I have a task for you. It's a simple task. Uh probably but not today. I'll start with it.

01:10:35

Vaibhav Singh: I talk to you tomorrow about it. Yeah. uh some things in mind. Okay. Okay. Okay, guys. I've added you to the group,

Avinash Patel: Okay.

Vaibhav Singh: Home Depot group, right? We I have added or not? I don't remember. Uh let me check.

Avinash Patel: Let me see.

Vaibhav Singh: Okay, I have not added. Let me add it. Okay. Okay, you've been added. Uh, let me just do a Okay. In the meantime, Samir, Fatima, someone, please help him with the ticket raising and whatever is required for on premises and the VPN things. All right. Please connect with them. Okay guys, uh thank you for joining. This is a supposed to be an half hour meeting. It's more than 1 hour. Thank you for your time. I'll talk to you tomorrow. In the meantime, if you have something, just post keep posting in the group. If there's any issues, post it in the group.

Avinash Patel: So,

Vaibhav Singh: Okay.

Avinash Patel: so all uh all whatever documentation uh PDFs code like that that's all going to

Vaibhav Singh: Oh yes, yes, yes, yes. Uh I have I have a zip for you.

Avinash Patel: be

Vaibhav Singh: So I can share but it's probably best I can I talk to you and explain to you what the zip is about that I'll do tomorrow not today. Uh Fatima uh can you share your thing and

Fatemeh Teimury: Yeah.

Vaibhav Singh: harish I think I can share there's a results document as well. I'll share that as well. Uh we have some documents. Let me repost it in the group. Okay you can go through it. Okay guys, bye. That's it. Thank you. Bye.

Sameer Shaik: Bye-bye.

Transcription ended after 01:12:49

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